Silent Burnout: A Multimodal AI Framework for Early Detection of Academic Disengagement

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***Abstract*—Leaving college is a soft process that cannot be spotted unless the student is disenfranchised by the course of study and the performance plummets drastically. This paper proposed a multi-model machine learning system which integrates the behavioral data of the students with the academic performance data to provide an indication of the probability of dropping out in the first time period that is possible. Data collection for the study has been done from the Open University Learning Analytics Dataset, which has more than 32,000 students in the database. The data has undergone feature engineering, and has been created to quantify the engagement of the students, the overall number of clicks, the number of days, and the mean number of interactions that the students have on a daily basis. This has been tested on several machine learning algorithms such as Logistic Regression, Random Forest, Gradient Boosting and XGBoost which showed a higher level of accuracy than the corresponding baseline algorithms, and the accuracy was 81.8.**

***Index Terms*—Student dropout, student disengagement, machine learning, learning analytics, educational data mining, ensemble learning, behavioral analysis, academic performance, Gradient Boosting, XGBoost.**

I. Introduction

As the number of online learning platforms and Learning Management Systems (LMS) increases, a steady stream of data on the interaction between students has been observed to flow constantly [12], [13]. Nonetheless, it has been one of the greatest challenges to determine the first signs of disengagement. Learners are likely to lose interest in learning activities eventually and then drop out of a course formally, therefore, it has been a very hard task to identify the initial manifestations of disengagement. Late disengagement detection implies that we cannot make our intervention as effective as possible and cannot assist the student who has become disengaged in time [1], [3].

Conventionally, identification of disengagement has been done on the grounds of academic performance like grades and test scores. These filters may be effective but they may not always depict behavior changes which were made by a student during disengagement. Majority of the symptoms at the initial stages of disengagement like decreased activity, unstable behavior and inability to engage on Learning Management System are not being monitored and, therefore, most disengagement symptoms are going undiagnosed [2], [5].

The ML methods can effectively overcome the shortcoming of the current study in examining the student data with the assistance of the big scale, behavior, and academic performance data to effectively forecast student disengagement prior to their identification by student academic performance. The literature on the use of EDM, in particular, the use of ML models to successfully predict risks of students, is also fairly extensive and has been performed in significant volumes of research work on the subject [2], [5]. Though the ML models have been successfully modeled so that they can effectively predict the risks of the students, it is largely done using the features of the academic students, and not using behavior features, except in cases where the academic features are used to establish the risks associated with the students, which is effective, as demonstrated in the studies cited in this paper, namely, [1] and [3]. Little progress is made in the use of advanced feature engineering techniques to well represent and predict student engagement. Train and refine the workings of your model by using data preprocessing, particularly by using feature engineering and scaling to achieve the most optimal results. When normalizing behavior (i.e. total clicks, active days, and frequency) you are literally normalizing the statistics on two or more scales so that these statistics are consistent across the same scale. By using a normalization technique, not only are you enhancing the stability of your models in such a way that you can effectively interpret the results of the application of your models, but you also are improving the results of your prediction more than ever in case you are using models that are highly sensitive to the distribution of the features it is using or relying on at the time that you are doing this itself would improve the stability of your models even more than this is already doing so

This project is aimed at establishing early disengagement warning signs in school and/or college which will be achieved through the behavioral engagement and academic performance

metrics as the key driving force in the formation of Multi-Modal Machine Learning Framework. Besides, it will enable you to implement an analysis of different models, including Logistic Regression, Random Forest, KNN, XGBoost, and Gradient Boosting, compare the performance of each of them in comparison with those discussed, and which one will provide you with the best outcomes in determining whether or not the student will become engaged or not.

The organization of the rest of this paper will be as follows: in Section II, we will be talking about the target data set, how features can be derived out of this data set, in Section III, we will be providing more details about various methodologies/models that will be used, in Section IV, we will be presenting our experimental results and finally in Section V, conclusions/possible future research will be presented.

II. Literature Survey

The issue of student disengagement has become a significant concern in the traditional and virtual learning environments. Previous studies were largely founded on academic performance, i.e., grades and examination outcomes, to determine the students who may be disengaged. Kotsiantis et al. suggested that machine learning methods could be used to distinguish students, although they conducted their study largely on academic performance, and behavioral aspects were not given much attention in their studies [1].

Due to the emergence of virtual learning, studies have been conducted on the role played by learning analytics and educational data mining in disengaging students. Romero and Ventura have suggested that Learning Management Systems produce quite a significant volume of data, which may be used to understand the behavior of students and discover concealed patterns [2]. They suggested that behavioral developments will be exhibited even before bad academic performance is noticed.

Many researchers have proven the effectiveness of behavioral indicators in detecting student disengagement at the early stages. Marbouti et al. discovered that students who reduce their degree of participation in online learning activities gradually tend to disengage than students who do not do so [3]. Macfadyen and Dawson discovered that clicks, active days, and submissions of assignments are powerful predictors of possible academic challenges to students [20]. This means that student behavior is better in identifying student disengagement than student grades.

Various machine learning algorithms are under implementation to enhance precision of prediction model. Logistic Regression is commonly used because it is simple and easy to interpret though it exhibits low accuracy level [10]. Conversely, the application of the Random Forest and K-Nearest Neighbors algorithm prove to be more useful in identifying student disengagement because the algorithms are capable of determining complex relationships between variables [6], [9]. Towards the recent past, there has been an increased demand in applying the Gradient Boosting and XGBoost algorithms in predicting student disengagement in learning institutions. This is because the algorithms can be able to detect patterns better than other algorithms can do so [7], [8].

Thus, the usefulness of academic and behavioral predictors in early identification of the disengagement of students has been demonstrated in the previous research.

III. Proposed System and Model Design

Figure 1 represents the system in question. Figure 1 presents us with the elements of the system and the data, how we obtain the features of the data and the machine learning algorithms are the primary elements of the system. The actions of processing the data are based on the data that pertains to the students and culminate to the risk prediction of the students being disengaged. These steps are applied to the data that we possess to create the system that will warn us in time in case the student is at risk of disengaging. The information, which we are utilizing in this project is the Open University Learning Analytics Dataset (OULAD), and this information is utilized very frequently in the studies concerning education [4], [15]. Three data sets of this data are being used by us i.e. studentInfo.csv, studentVle.csv and studentAssessment.csv. The data gives us the information related to the students such as the details of the students, such as who they are and how they behave in the Learning Management System and the results they have gained in the tests. The data includes the data concerning 32,000 students. There are over 10 million records in the data on the occasions, the students have used the Learning Management System, and this is such a good amount of data particularly to see how the students behave.

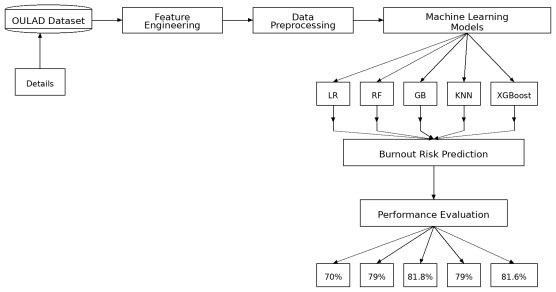

Fig. 1. Proposed System Models

StudentInfo file contains the data on students, i.e. where they belong, what courses they pursue, etc. StudentVle file contains data on the use of the learning system by the students i.e. what the students are clicking on, etc. The studentAssessment file contains grades and other data of the school work of the students. This information as a combination will provide us with an insight on the behavior and performance of students in school. This is needed to know whether the students are not involved in the system of learning or not [4], [20].

It is this data that is being used to ascertain the probability of students becoming disengaged rather than whether they will fail. Our system has developed an indicator that indicates whether the students are Low Risk, Medium Risk and High

Risk of disengaging with the learning system. This is achieved through observing the way students are behaving and doing at school. This will enable us to target those students who are gradually becoming out of interest, rather than those who are already having trouble. When the system is being prepared, it receives the information in the raw files. The system relies on the data out of the raw data i.e. clicks, active days, average number of clicks per day, average score, assessments submitted, standard deviation of scores etc. This will provide an idea of the level at which the students are taking part in the learning system as well as their level of performance at school. The system includes numbers that are continuously changing and the numbers that are summed up and this is what makes the system good in determining what is important, choosing the features and looking at the data in order to know what is going on in the data [14], [20]. The system will bring together the combination of the three data files which the system will be fed on through the name of the student and the details of the class. The data is then prepared to be used by the system by processing the data, eliminating duplicates, verifying the data against errors and ensuring that all the numbers are normal and in case they are to appear on the scale they are. When the data lacks information, the system replaces the information by the missing data. The system then generates features that indicate the engagement levels and the consistency levels of the students. The system then uses the features to produce the features that the system will be using to do the predictions.

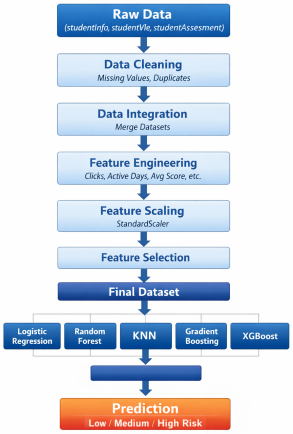

Fig. 2. Data processing pipeline for student risk prediction

Figure 2 illustrates the entire data processing process of the proposed system. The data processing pipeline comprises of data cleaning, data integration, feature engineering, scaling, and feature selection. The data processing workflow is shown in the next figure.

There exists a feature selection in the context that is to be offered. Here we receive the useful information and dispose of the things not so useful and repeaters. This will enable the model to be more efficient in what it does and eliminate the noise. It will also make it more competent in things that it has never observed. Learners become inactive due to actions. Due to school we select features that will simplify the system to use because of school we do so [10], [14].

The whole system is made up of four parts, that is, the part in which we feed the data, the part in which we manipulate the data, the part in which we model the data and the part in which we make predictions. In the part where we input the data, we input all the data concerning the ways in which the students are interacting with the learning system, the personal details of the students, and the grades of the students [12], [13]. Then in the stage where we process the data we clean the data, we assemble the data and we create the features [10]. Then, during the section where we train the model, we train the computer with a variety of methods, that is, Logistic Regression, Random Forest, Gradient Boosting, and XGBoost, on the cleaned-up data. Finally, in the part where we make predictions, we make predictions using the model concerning the students who are getting disengaged using the model. The whole system is created in a way that we can make predictions about how the students are becoming disengaged in classes in an easy and comprehensible way and in a way that we can make predictions about a large number of students.

IV. Classification Algorithms

We are attempting to know how to categorize, or to put information in groups that already exist. The primary purpose of classification is to arrive at an educated guess of what thing belongs to what group when we are not certain. This is what we are doing by considering the trends in the information that we already possess [10], [2]. We would like to categorize the data to determine whether the students are Low Risk, Medium Risk or High Risk of dropping out of school. We are examining how the students act and what they are doing in school to make an educated reasoning as to which group they belong to [14]. This information is coming through the computer system of the school.

The fundamentals as to how we are doing this are as follows.

A. Logistic Regression

In this paper, we apply a technique known as the Logistic Regression to categorize items. It implies that we split our data in two boats, the training one, and the testing one, to determine the success of our model. This is the classification problem that is being used and it is referred to as Logistic Regression and is generally simple to comprehend and utilize, and it is also likely to work well in such cases. It is a mathematical calculation that defines the probability of a piece of data to belong to a particular group, depending on the features that are available in the data. Even though a machine learning model, when introducing a classification problem, a good starting point would be a Logistic Regression. In this research, we are going to classify our data (in this case, our students) into various categories (using the Logistic Regression) based on the manner in which they are to disengage using various features like academic and behavioral features. This has been a successful study and we were able to classify our data 70

percent of the time, which is quite good to compare to other machine learning model, which are more complex. However, a machine learning model, when performing a classification problem, one can start with a Logistic Regression and this study has demonstrated how successful this model can be, a success rate of 70 percent, a good parameter to compare to other more complex models, and a machine learning model, such as this one.

The mathematical formula of this operation can be stated as the following:

$$F(x) = \frac{1}{1 + e^{-x}} \tag{1}$$

Where:
- x is a weighted feature sum.
- e is a constant.
- $F(x)$ is projected value, a probability ranging between 0 and 1.

B. Random Forest

The concept of random forest is a machine used in categorizing items by creating many decision trees and combining their results to obtain the answer. When we apply the Random Forest system, then we take the information we have and repeat it many times with a different set of things to consider each time but then look at what they all think and go with what they all say the most. This system has the advantage of handling non-linear relationships and ensures that the model does not get too close to the training data and the entire system becomes more stable in the process of doing so [6]. We therefore applied the Random Forest system which groups the students based on their characteristics with regard to the disengagement aspect. The system is good at locating such complicated relationships since the process of disengaging the students who are doing something can be dependent on many factors such as what they are clicking on when they are doing something and their performance on the test among other factors [14]. Random Forest model performed better in comparison to the Logistic Regression model and obtained the answer right 79 percent of the times.

Random Forest has a prediction ability that can be formulated as follows:

$$\hat{y} = \text{majority_vote}(T_1, T_2, \ldots, T_n) \tag{2}$$

Where:
- T 1, T 2, and so on, T n are the random forest decision trees.

C. Gradient Boosting

Gradient Boosting is a learning technique that involves a multitude of models in order to create one strong model. The models are not complex and one model follows another, one model at a time. The purpose of constructing one model after another is to make sure that the errors of the one model are rectified by the other model. This will be more accurate in the predictions and errors are minimized. It is quite effective in establishing what group thing belongs to, particularly where there is a complex connection between things that characterize what that group is and what group that thing belongs to [7]. It has been utilized in this project since there are many reasons why students stop paying attention in the classroom and some of them arise in school, others arise out of school and some may be complex reasons. The algorithm relies on numerous simple models, successively, to create a single strong model. Gradient Boosting was the best of all the models that were used in this study particularly in identifying what group something belonged to where it was able to get it correct 81.8 percent of the time.

The predictive role which forecasts the functioning of things in Gradient Boosting may be specified as:

$$F_m(x) = F_{m-1}(x) + \gamma_m h_m(x) \tag{3}$$

where:
- $F_m(x)$ is the revised prediction model in stage m.
- $F_{m-1}(x)$ is the prediction of the last stage.
- $h_m(x)$ is the weak learner at level number m.
- γ_m the learning rate or weight of the weak learner.

D. K-Nearest Neighbors

The k-nearest neighbors algorithm also known as KNN is a technique of classifying objects. It examines all the data points that are closest to the data point it is attempting to classify and it then classifies it in terms of what the majority of those data points share. It does not attempt to know how things work, it only attempts to remember what it has seen previously. It applies this approach to come up with its predictions. KNN algorithm is very simple. It simply attempts to retain all the information that it has previously noticed and some subsequent information is then utilized as it tries to make its predictions.

The KNN algorithm is based on a factor known as k which is basically the number of nearest neighbors that we want to take into consideration. In our research, we used the KNN algorithm where we tried to see who amongst the students would be at risk of becoming disengaged with school. This we did by comparing the similarity of a student to others on the basis of the level of engagement and the level of academic performance that they had taken [14]. We had done all the cleaning of the data and ensured that everything was in the range so that the KNN algorithm could pick up some of the patterns in the data that could not be seen at first glance. KNN algorithm could identify the student who was at risk of becoming disengaged at approximately 79 percent.

The following rule can be formulated as used by the KNN algorithm in making a prediction:

$$\hat{y} = \arg \max_{c_j \in \mathcal{N}_k(x)} \sum_i I(y_i = c_j) \tag{4}$$

where:
- $N_k(x)$ is a group of k closest neighbours to a certain data point x.
- $I(y_i = c_j)$ is an indicator function equal to 1 when the neighbor belongs to class c_j, and 0 otherwise.
- $\hat{y}$ is a class prediction.

E. XGBoost

XGBoost or Extreme Gradient Boosting is a nice algorithm, which will assist us in making the right decision in a short period of time. XGBoost is a variant of the Gradient Boosting model and it is commonly applied to classify data. XGBoost operates by building decision trees sequentially and each new trees is expected to correct errors of the other trees. XGBoost also operates on the basis of several rules that prevent the model to become excessively good at the task and enables it to make good predictions even with new information. The XGBoost is applied to identify the students at risk of becoming disengaged at school, depending on the relationship between the behavior and performance. The XGBoost model could make such a prediction 81.6 percent of the time, which is near optimum possible prediction, which is attained by the Gradient Boosting model.

The main objective of XGBoost can be described with the help of an equation:

$$\mathcal{L}^{(t)} = \sum_{i=1}^{n} l\left(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)\right) + \Omega(f_t) \tag{5}$$

where:

- l is a loss statistic that quantifies the actual and predicted values.
- $f_t(x_i)$ is a prediction of the tree added at iteration t.
- $\Omega(f_t)$ is a regularization term.
- $\mathcal{L}^{(t)}$ is an objective which is minimized at step t.

V. RESULTS AND DISCUSSION

To test the proposed machine learning model, it was tested to assess its performance. The evaluation was done by using the disengagement data set, which was cleaned up and then fed into the model using different classification algorithms. The data set was divided into two, one of which was used to train the model and the other to test the performance of the model. The performance was measured using evaluation metrics, and the model's performance was determined using the evaluation metrics. Table I demonstrates the assessment measures of the study.

All these assist us in analyzing the model working performance and in forecasting and pointing out the students that are at risk of engaging.

TABLE I
EVALUATION METRICS USED IN THE STUDY

Metric	**Description**	**Importance**
Accuracy	Proportion of accurate predictions to total predictions	Assesses the model performance
Precision	Ratio of correctly predicted positives of total predicted positives	Denotes whether the model is right or not about making predictions
Recall	Ratio of predicted positives correctly prediction to actual positives	Denotes whether the model is able to identify the riskiest students or not
F1-Score	Mean of precision and recall that are harmonic to each other	Gives the tradeoff between accuracy and recall

The findings of the experiment indicate the performance of various algorithms, which are either good or poor in predicting student disengagement. The Logistic Regression model was used as a baseline. This implies that it was a model against which we could compare all other models. It was able to get 70 percent of its predictions right. Although this model was quite simple and easy to comprehend, it could not work effectively the same way as some other models that incorporated various approaches. It is thus reasonable to say that this correlation between the level of engagement of a student and his/her performance at school is complex, and although this model was easy to use and understand, it could not do well considering the performance of other models. The machine learning architecture and the prediction of student disengagement is rather important in comprehending the level of effectiveness of this model.

Random Forest approach was capable of doing much better than the Logistic Regression approach. It was capable of making it correct 79 percent of the time. It could do better than the Logistic Regression methodology, as it is a blend of decision trees, and this prevents its over-performance in its job [6]. It can thus work better with things that are non-linear such as the number of times a student clicks, the number of days the student is active, average score of the student, and even the way the student submits his work. Even the K-Nearest Neighbors algorithm could score it correctly 79 percent of the time, and this is to say that it was capable of doing well since it could observe similar trends in the interaction and engagement of students and how they could possibly achieve disengagement.

Among the techniques we experimented upon, the Gradient Boosting method was the best. This approach was correct on 81.8 percent of occasions. This approach was then followed by the XGBoost approach that estimated it at about 81.6 percent accuracy. This demonstrates that the techniques that apply the boosting technique are effective in solving this problem since they are always in search of the better the model to be and identify patterns in the data that are not easily noticeable [7], [8]. The more complex methods which utilize a body of models are superior to the simple methods since the students are losing interest because of the factors which interact with one another such as the behavior of the student and the academic performance of the students.

The performance of the model is given in a comparative manner in Table II.

TABLE II
PERFORMANCE OF CLASSIFICATION MODELS

Model	Accuracy (%)
Logistic Regression	70
Random Forest	79
K-Nearest Neighbors	79
Gradient Boosting	81.8
XGBoost	81.6

We obtained very clear results. They demonstrate that the Ensemble methods are faring much better than the Logistic

Regression model. This indicates that the issue of determining when the students will disengage is not a straightforward issue and therefore, cannot be easily resolved. There are patterns and interactions which are at play here. The tree based boosting algorithms are more suitable to these patterns [7], [8].

In the project, Gradient Boosting worked a little better than XGBoost. Thus, it was the algorithm of all models that we have tried.

The experiments that we carried out taught us one important thing. The features that present us with the way the students act are extremely significant in the process of forecasting the likelihood of the students dropping out of school. The characteristics that are the number of clicks, the number of days that the student has been active and the average number of clicks per day are quite good indicators that the student has the potential of losing interest in school [14], [20]. This is more so when we are pooling these characteristics with such characteristics as the mean of the score, the number of tests completed, and the standard deviation of the score. This informs us that, in our examination of the features of the students, we should not simply look at the features which reveal us how the student is fairing academically when we are attempting to establish the reasons why students lose interest in school. Rather we should consider the features that present us with how students are acting and the features that present us with how students are performing in school to understand the cause of students disengaging in school. The two aforementioned ensemble learning techniques are quite excellent in terms of estimating the risks of students dropping out of school since they are capable of addressing all these characteristics [7], [8].

In general, the experiments demonstrate that the multimodal machine learning method is actually effective in early detection of disengagement. The two ensemble models, Gradient Boosting and XGBoost, were quite effective. This implies that advanced machine learning methods can be used to provide viable solutions to the prediction issue, particularly when used in online learning, like the one that relies on Learning Management Systems [2], [12].

VI. CONCLUSION

The current paper is on one of the methods of identifying when students are becoming uninterested in school. We accessed a computer system that indicated the behavior of students and their performance in school. The computer system at schools provided this information to us [12], [13]. We did not just want to know the results of the students at the end of the year but rather those that are slowly losing interest. We examined such things as how many times students clicked on things how many days they were active how many clicks they made a day how many tests they have taken and how their scores have changed. These were used to create a model of a computer that can tell when students are getting bored [14], [20].

We experimented with some computer models such as Logistic Regression, Random Forest, Gradient Boosting, K-Nearest Neighbors and XGBoost. These models were compared to determine which one was the best. The models that predicted using different things performed better as compared to the simple Logistic Regression model. The optimal Gradient Boosting hit 81.8 percent. XGBoost made the correct prediction concerning 81.6 percent of the time. This demonstrates that the timing of loss of interest in students is not easily predictable and requires a computer model such as Gradient Boosting or XGBoost to be accurate. The prediction of academic disengagement relies on the behavioral and academic characteristics being good and the models of Gradient Boosting and XGBoost being good at predicting this [7], [8].

According to the study, the indicators of behavioral engagement can indeed be helpful in determining whether a student is at risk of disengagement at an early age or not [14], [20]. The system which we developed can classify students into three categories i.e. Risk, Medium Risk and High Risk. That is why it is an excellent tool to alert us in the case when a student is in danger of being disengaged. The research confirms that machine learning is one method of detecting student disengagement in online learning [2]. Machine learning can be used to address the problem of student disengagement. The paradigm is a feasible remedy, in online education and student inattention.

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